Significant Figures and Measurement

Counting significant figures, scientific notation, rounding and calculations

Name _____ Class _____ Date _____

RULES

Leading zeros: not significant · Zeros between digits: significant · Trailing zeros after a decimal point: significant

× and ÷: fewest significant figures · + and −: fewest decimal places · exact numbers do not limit the answer

Rounding an exact 5: round to the even digit (2.345 → 2.34; 2.355 → 2.36)

$$\bar{x} = \frac{\sum x_i}{n} \cdot s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \cdot \text{RSD} = \frac{s}{\bar{x}} \times 100\%$$

Give every answer with its unit.

A. Understanding

1. A ruler is marked in millimetres. Explain why a length measured with it can be recorded as 4.37 cm but not as 4.372 cm.

2. Explain the difference between accuracy and precision. Why can't precision be judged from a single measurement? Can a set of measurements be precise but not accurate? Give an example.

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B. Counting significant figures

3. How many significant figures are in each? (a) 0.0050 m (b) 30.70 g (c) 6.022×10^{23} mol^{-1} (d) 0.10200 L (e) 1200 mL

C. Scientific notation and rounding

4. Write in scientific notation, keeping the significant figures: (a) 0.000 350 m (b) 45 600 g (to 3 significant figures) (c) 0.0207 L

5. Round each to three significant figures: (a) 2.4562 g (b) 0.0012349 m (c) 18 750 s (d) 9.9951 mL (e) 4.365 g (f) 4.375 g

D. Calculations

6. Calculate and give each answer to the correct number of significant figures: (a) 3.25 cm × 2.1 cm (b) 100.0 g ÷ 3.0 mL (c) 8.315 J/(mol·K) × 298 K

7. Calculate to the correct precision: (a) 12.0 g + 0.335 g + 4.5 g (b) 6.38 cm − 2.1 cm

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E. Challenge

8. An aluminium block measures 2.00 cm × 3.00 cm × 4.00 cm and has a mass of 64.8 g. Calculate its volume and its density, each to the correct number of significant figures.

9. A student weighs a certified 2.500 g standard mass in triplicate and records 2.52 g, 2.48 g and 2.51 g. (a) Calculate the mean, the standard deviation and the relative standard deviation. (b) Calculate the error and the percent error. (c) Comment on the accuracy and precision, and report the result as mean ± standard deviation.

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